

Question number	Answer	Additional guidance	Mark
6(a)	An answer that includes the following points: a {chemical / substance / drug / medicine} that {kills / inhibits the growth of} {bacteria / microorganisms / pathogens} (1)	ACCEPT inhibits the reproduction of microorganisms chemical that is bacteriostatic / bactericidal DO NOT ACCEPT confusion with antigens / antibodies viruses	(1)

Question number	Answer	Mark
6(b)	The only correct answer is C <i>A is incorrect because $((10 - 1.27) \div 1.27) \times 100 = 687.40157$</i> <i>B is incorrect because $((10 - 1.27) \div 1.27) \times 100 = 687.40157$</i> <i>D is incorrect because $((10 - 1.27) \div 1.27) \times 100 = 687.40157$</i>	(1)

Question number	Answer	Additional guidance	Mark
6(c)	An explanation that includes two of the following points: - bacteria will not be resistant to {this / new} drug (1) - so we will be able to treat {antibiotic-resistant bacteria / (bacterial) diseases / (bacterial) infections} (1) - if the new drug is not an antibiotic, bacteria may not develop resistance (1)	ACCEPT pathogens throughout DO NOT ACCEPT viruses ACCEPT to win the evolutionary race current drugs are less effective IGNORE bacteria are becoming resistant to current antibiotics ACCEPT so fewer people will die from bacterial infections if we do not have a new drug, we will not be able to treat bacterial infections	(2)

Question number	Answer	Additional guidance	Mark
6(d)	An answer that includes the following points: • drug {disrupted / increased permeability of} {membrane / wall / peptidoglycan} (1) • so antibiotic could {access / affect} their {target sites / named target sites} / so antibiotic can enter cell (1)	ACCEPT acts as a channel protein e.g. peptidoglycan, ribosomes, enzymes, cell wall, murein ACCEPT less antibiotic needed to disrupt {cell wall / peptidoglycan} as drug has done this	(2)

Question number	Answer	Additional guidance	Mark
6(e)(i)	A description that includes two of the following points: • (overall) all three antibiotics decrease the species richness (1) • effect of all three antibiotics is (nearly) the same in the first 3 days (1) • antibiotic C {decreases species richness the most / destroys most bacteria / is the most effective at killing bacteria (overall / after 3 days)} (1)	ACCEPT it / number of species / richness ACCEPT antibiotics only show different effects after 3 days ACCEPT converse for antibiotic A	(2)

Question number	Answer	Additional guidance	Mark
6(e)(ii)	An explanation that includes three of the following points: • {antibiotic / C} may act {faster / at lower doses} (1) • therefore the {species richness / gut flora} may not {decrease / be affected} as much (1) • credit appropriate comment linking species richness to health (1) • patient may recover sooner from their infection (1)	ACCEPT C can be taken for fewer days e.g. gut flora will be able to compete with pathogens for {space / nutrients} less damage to gut flora will aid digestion ACCEPT fewer side effects	(3)

